Python & Traditional NLP with NLTK

Cleaning raw news articles and performing sentiment analysis using Natural Language Processing techniques

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What is Traditional NLP?

Natural Language Processing

Computers understanding human language through computational methods

Rule-Based Methods

Statistical models and linguistic rules instead of deep learning

Key Libraries

NLTK, spaCy, and TextBlob for text processing and analysis

Phase Objective

Build a complete pipeline to process raw news text: clean it, tokenize into meaningful units, remove noise, normalize words, and analyze sentiment using VADER.

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Introducing NLTK

Natural Language Toolkit

A powerful Python library designed for research and education in computational linguistics. NLTK provides easy-to-use interfaces for over 50 corpora and lexical resources.

Core Features

- Sentence and word tokenization
- Stopword removal
- Stemming and lemmatization
- Part-of-speech tagging
- Sentiment analysis with VADER

Installation

```
pip install nltk

import nltk
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('vader_lexicon')
```

Download required datasets once to use all NLTK features.

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Raw News Article Example

Sample News Text

"The Indian stock market crashed today due to global economic concerns. Investors are worried about inflation and rising interest rates. However, some experts believe the market may recover soon."

Our Processing Pipeline

01

Clean Text

Remove special characters and normalize formatting

02

Sentence Tokenization

Split into individual sentences

03

Word Tokenization

Break sentences into words

04

Stopword Removal

Filter common words

05

Lemmatization

Normalize to base forms

06

Sentiment Analysis

Calculate emotional tone

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Sentence Tokenization

Splitting Text into Sentences

Sentence tokenization divides continuous text into discrete sentences using punctuation and contextual rules.

```
from nltk.tokenize import sent_tokenize

sentences = sent_tokenize(news_text)
print(sentences)
```

Output Result

```
[ 'The Indian stock market crashed today due to global economic concerns.',
 'Investors are worried about inflation and rising interest rates.',
 'However, some experts believe the market may recover soon.' ]
```

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Word Tokenization

1

Input Text

Complete news article as string

2

Tokenization

Split by spaces and punctuation

3

Word List

Individual tokens in order

```
from nltk.tokenize import word_tokenize

words = word_tokenize(news_text)
print(words)

# Output:
['The', 'Indian', 'stock', 'market',
'crashed', 'today', 'due', 'to',
'global', 'economic', 'concerns',
'.', 'Investors', 'are', 'worried',
'about', 'inflation', 'and', 'rising',
'interest', 'rates', '.', 'However',
'some', 'experts', 'believe', 'the',
'market', 'may', 'recover', 'soon', '.']
```

Key Points

- Punctuation marks become separate tokens
- Case is preserved in original form
- Words maintain their sequential order
- Ready for further processing steps

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Removing Stopwords

Filtering Common Words

Stopwords are frequently occurring words that carry little semantic meaning. Removing them reduces noise and focuses analysis on meaningful content.

1

Common Stopwords

is, the, and, to, are, of, in, that, it, for, with, as, on, at, by, this, from, but, not, or, be, have, has, had, been

2

Implementation

```
from nltk.corpus import stopwords

stop_words = set(stopwords.words('english'))
filtered_words = [w for w in words if w.lower() not in stop_words and w.isalpha()]
print(filtered_words)
```

3

Result

Significantly reduced token count focusing only on meaningful words relevant to sentiment analysis

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Lemmatization

Converting to Base Forms

Lemmatization reduces words to their dictionary base forms using morphological analysis. Unlike stemming, lemmatization ensures valid words.

Examples

- crashed → crash
- rising → rise
- investors → investor
- concerns → concern
- rates → rate

```
from nltk.stem import WordNetLemmatizer

lemmatizer = WordNetLemmatizer()
lemmatized_words = [lemmatizer.lemmatize(w.lower()) for w in filtered_words]
print(lemmatized_words)
```

Benefits

Normalizes different word forms into single representation, improving text analysis accuracy and reducing vocabulary size.

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VADER Sentiment Analysis

Valence Aware Dictionary

Pre-built lexicon of words with emotional intensity scores

News & Social Media

Optimized for short, informal texts with emoticons and slang

Four Scores

Positive, negative, neutral percentages plus compound score from -1 to +1

Implementation

```
from nltk.sentiment import SentimentIntensityAnalyzer

sia = SentimentIntensityAnalyzer()
scores = sia.polarity_scores(news_text)
print(scores)

# Output:
{ 'neg': 0.237, 'neu': 0.608, 'pos': 0.155, 'compound': -0.1280 }
```

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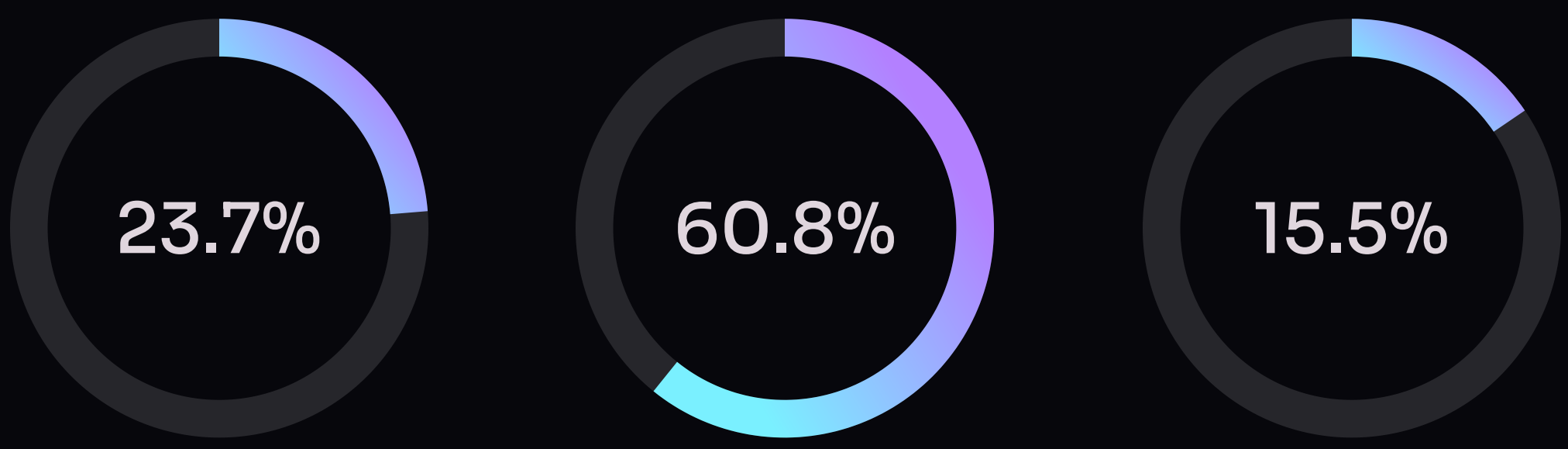
Complete Pipeline & Results

Integrated Code

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from nltk.sentiment import SentimentIntensityAnalyzer

news_text = """The Indian stock market crashed today due to global economic concerns. Investors are worried about
inflation and rising interest rates. However, some experts believe the market may recover soon."""

# Tokenization
words = word_tokenize(news_text)

# Stopword Removal
stop_words = set(stopwords.words('english'))
filtered = [w for w in words if w.lower() not in stop_words and w.isalpha()]

# Lemmatization
lemmatizer = WordNetLemmatizer()
lemmatized = [lemmatizer.lemmatize(w.lower()) for w in filtered]
print("Cleaned Words:", lemmatized)

# Sentiment Analysis
sia = SentimentIntensityAnalyzer()
sentiment = sia.polarity_scores(news_text)
print("Sentiment Scores:", sentiment)
```

Interpreting Results



Negative
Concern and worry terms

Neutral
Factual reporting

Positive
Recovery hope

Compound Score: -0.128

Negative overall sentiment since compound score is less than 0

Learning Outcomes

Text Cleaning Pipeline

Complete workflow from raw text to processed tokens

NLTK Proficiency

Master tokenization, stopword removal, and lemmatization

Sentiment Analysis

Apply VADER to quantify emotional tone in news and social media

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